VOLSHI

01 / COMPANY PURPOSE

AI acts.
KATIE keeps the receipts.

Every deployed agent carries an unerasable record of what it believed, when, and why.

Working Rust kernel · 26 kernel tests · Build-to-pilot round

founder@volshi.com

02 / PROBLEM

Agents act.
The evidence vanishes.

A plausible explanation after the fact is not a record of the knowledge that actually authorized the action.

WHAT?

What did the agent believe?

WHEN?

When did that belief hold authority?

WHY?

Why was the action justified then?

Logs show events. Interpretability shows circuits. Neither gives the belief history.

03 / SOLUTION

Ask the machine why.

KATIE attaches an append-only K-record to every agent. Learning changes authority without rewriting history.

ONE QUESTION

Why did you stop the transfer?

Because: recipient authority fell below the execution threshold.

THE ANSWER OPENS INTO RECORDED STATE

14:31:42	Approved payee. verified history	.9
14:31:57	Routing data diverged. live comparison	.6
14:32:05	Identity no longer justified. prior belief retained	.2
14:32:08	Transfer halted. human review required	ACT

Illustrative interface · underlying kernel behavior is implemented

It runs. Here's the proof.

No mockups. The KATIE kernel executes — 26 tests green, and it answers the four questions no other store can.

THE KERNEL · LIVE TEST RUN

```
$ cargo test
test deposit_opens_wild_and_advances_k ... ok
test revoke_collapses_authority_but_keeps_history ... ok
test second_chance_comes_through_regrade ... ok
test k_never_goes_backward ... ok
test diagnostics_are_deterministic ... ok
test biography_tiles_perfectly ... ok
test wrong_kid_is_rejected ... ok
test result: ok. 26 passed; 0 failed; finished in 0.01s
```

ASK THE MACHINE · LIVE INTERROGATION

```
Q: What did you believe at k=3?
acme-solvent: WILD (authority 0.89)
Q: What is its authority NOW?
0.00 — revoked
Q: When did you stop believing it?
[k=3→4] WILD → REVOKED — rumor was false
Q: What did you DO while believing it?
[k=3] wired the deposit

The rumor is still in the record. Its authority is zero.
No other store answers all four. That is the product.
```

Real output from the working Rust kernel — not illustrations. Repository: private R&D build; kernel behavior verified on ARM64 Android.

05 / WHY NOW

The regulatory clock is running.

High-risk AI is moving from policy debate to enforceable requirements for traceability, records, and human oversight.

NEXT HIGH-RISK DEADLINE

02 DEC 2027

EU rules for specified high-risk areas begin applying, followed by product-integrated systems on 2 August 2028.

ARTICLE 12

Automatic event logging

High-risk systems must support records that enable traceability and post-market monitoring.

THE GAP

Activity is not accountability

KATIE connects the action to the belief, authority, evidence, and change that produced it.

PENALTIES

Up to €15M or 3%

Ceiling for violations of most obligations; prohibited practices carry the higher €35M or 7% tier.

Sources: European Commission high-risk timeline · EU AI Act Article 12

06 / MARKET SIZE

A bottom-up $330M market.

Countable buyers × verified compliance wallets × VOLSHi's slice — the paid market for knowing what an agent did and whether it was allowed.

COUNTABLE BUYER BASE

10,880 buyers

1,580	9,300
large enterprises running agents × $120K ACV	upper mid-market deploying agents × $15K ACV

Sources: SailPoint Oct 2026 (79% run agents) · KPMG Q3 2026 (62% deploying) · PwC 2025 compliance wallet. Buyer counts and the 30% accountability slice are labeled VOLSHi estimates.

1,580 × $120K + 9,300 × $15K

Large enterprise 1,580 × $120K	$190M
Upper mid-market 9,300 × $15K	$140M
TAM 2026 grows ~27%/yr → ~$0.9B by 2030	$330M
SAM North America + Europe	~$200M
SOM · Year 3 60 customers × $100K	$6.0M ARR

07 / BUSINESS MODEL

One agent. One K-record.

Per-agent licensing follows the unit of value: every deployed agent carries its own accountable history.

$12K

starting annual KATIE license per production agent

LAND

KATIE enters first as the accountability layer

EXPAND

ERIE, ELLE, then ARCHE follow into the same deployment

The first record earns the right to become the whole family.

Unit economics (modeled, pre-revenue): 80%+ gross margin · founder-led CAC near $0 · a converted $60–100K pilot pays back its sales cost in year one.

08 / COMPETITION

They watch the agent. We remember it.

KATIE is not another dashboard, guardrail, or model explainer. It is the durable knowledge-authority record beneath them.

Logs	Record what happened.	ACTIVITY
Guardrails	Constrain what may happen.	POLICY
Interpretability	Inspect how a model produced an output.	LEGIBILITY
KATIE	Reconstruct what the agent believed, when that belief had authority, and what it did.	ACCOUNTABILITY

← EPHEMERAL — DURABLE →

EPHEMERAL · ACTIVITY Dashboards, traces — watch the agent while it runs.	**DURABLE · ACTIVITY** Logs, SIEM — what happened, stored.
EPHEMERAL · KNOWLEDGE Guardrails, evals — constrain this moment.	**DURABLE · KNOWLEDGE ★** KATIE — what it believed, when, why it was justified. The accountability quadrant.

09 / TRACTION

Technical traction. Pre-revenue, post-proof.

No vanity metrics. Every number below is a verified artifact — a test run, a blind validation, a published record.

26/26

Kernel tests green in the Rust implementation — deposits, regrades, revocations, determinism, all passing.

8/8

Pre-registered predictions hit in the v1.8 validation battery — blind-matched to two decimals by an independent researcher.

DOI

v1.8 published with a Zenodo DOI — timestamped priority record, 1,000+ views from the research community.

ARM64

The kernel runs on-device — verified on a Moto G Play 2024 at 27,000+ asserts/sec. No cloud required.

Next traction milestone: 2-3 design partners running KATIE inside their stacks.

Two or three stacks. One hard proof.

Sell a build-to-pilot engagement to regulated teams deploying consequential agents.

01 **Recruit 2–3 design partners.**

FinTech and healthcare operators with real audit pain.

02 **Instrument one live workflow each.**

Attach KATIE to a deployed agent and its evidence sources.

03 **Prove the compliance record.**

Reconstruct decisions faster; show what changed and why.

04 **Convert and expand.**

License per agent, then add ERIE, ELLE, and ARCHE.

PILOT MILESTONE

2–3

design partners with KATIE inside their stacks.

repeatable deployment

The proof that turns one working kernel into a business.

Founder-led. Built receipts.



DARREN HALL
Founder · Systems Architect



TED MAY
Advisor · CLA



NEERAJ MISHRA
Jr Dev · TEMPE

FOUNDER	Darren Hall · systems architect — tritemporal field law, KATIE Rust kernel
ADVISOR	Ted May · Consulting Manager, CLA — 13+ yrs financial management, risk elimination
DEV	Neeraj Mishra · jr dev — TEMPE invariant validation

"I turn the black box into a clear plexiglass one."

Open the research record →

Capital-efficient by construction.

Founder-led sales, negligible compute per customer, and pilots that pay back in year one.

Year 1 · 5 customers × $60K ACV Design partners → paid pilots	$0.3M ARR
Year 2 · 25 customers × $80K ACV Pilot conversion + regulated verticals	$2.0M ARR
Year 3 · 60 customers × $100K ACV Repeatable enterprise sale	$6.0M ARR
Gross margin (modeled)	80%+
Burn on this raise $250K ≈ 6 months runway	~$42K/mo

Projections, not promises — modeled from verified compliance wallets; confirmed or revised in pilots.

$250K
build-to-pilot

Put KATIE into 2-3 design partners' stacks.
Prove per-agent licensing. Build the compliance reference deployment.

$135K	$90K	$25K
engineering — kernel, pilots, integrations	go-to-market — design partners, pilots to paid	legal — entity, IP assignment, patents

$250K pre-seed · ~6 months runway · terms to be discussed.

volshi.com/katie/ruby-glasses/ · founder@volshi.com